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| Course   | : Diploma in Infocomm and Media Engineering (EGDF21)               |
| Module   | : Cross Platform Mobile Application Development (EGL303)           |
| Project  | : Development of a Cross Platform Mobile Application using Flutter |
| Software | : Visual Studio Code / Flutter & Dart SDK / Android Studio         |

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### 1. Objective

This mini-project requires you to put into practice the application design concepts that you have learned over the duration of this module. Specifically, you will need to construct a Flutter Mobile Application. You need to demonstrate the workability of your application on emulator or actual device.

You may propose any application title that complies with the theme for the semester.

### 2. Project Theme

The project theme for this semester is "**My Smart City: Connecting Our Senses**".

### 3. Background

Singapore strives to become a Smart Nation to support better living, stronger communities, and create more opportunities, for all. The Smart Nation initiative is one where people are empowered by technology to lead meaningful and fulfilled lives. Through harnessing the power of networks, data and info-comm technologies, it seeks to improve living, create digital economic opportunities, and build a closer community.

The Smart Nation and Digital Government Office in the Prime Minister's Office, supported by other government agencies, coordinate the Smart Nation initiative. The team has identified 5 key domains that will have a significant impact on the citizen and society, and they are:

- ❶ Transport
- ❷ Home & Environment
- ❸ Business Productivity
- ❹ Health and Enabled Ageing
- ❺ Food & Beverage

You may propose a project to serve any (but not restricted) of the above domains. You are required to gather the relevant information from various sources, plan, propose, design and develop a smartphone app that can bring convenience to the Singapore Citizen/Society.

### 4. Minimum Project Requirements

**4.1 [LOGIN/REGISTRATION](#)** When the application is launched for the first time, the [Login page](#) will be presented to login and authenticate user. Provision of a link to [Registration page](#) for new user to register their *email ID* and *password*.

- 4.2 PROFILE** The application should capture and store user's particular (locally store in [shared preference](#) or remotely store in [back-end server](#), e.g., [Firebase/FireStore/Storage](#)). On the subsequent time that the application is started, the application will just display the greeting to user. Menu item should be provided to allow user to change his/her profile.
- 4.3 ABOUT** The application should display brief description about the application, company, contact information and its developer. It should provide the link to call company directly and email feedback to company.
- 4.4** To create application pages with [functional features](#) (**minimum 4 functional pages** excluding *Login, Profile & About pages*) using Tabbed, Drawer or any other navigation widgets, and apply many other types of Flutter widgets, such as ListView, GridView, etc., in your UI design.
- 4.5** All data MUST BE stored in back-end server database, e.g., Firestore/Storage. You can use/create any possible data sources (e.g., JSON Dataset) that will help to meet the intended objectives of the app. Some useful data sources are as followed:
- Data.gov.sg - <https://developers.data.gov.sg>
  - LTA Mall - <https://www.mytransport.sg/content/mytransport/home/dataMall.html>
  - Tourism Info Hub - <https://tih-dev.stb.gov.sg/getting-started>
- 4.6** The user interface should conform to Android or iOS mobile standards, which should be intuitive and mobile user friendly.

## 5. Project Criteria

- 5.1** This is an **individual** project. If you are caught plagiarizing (copying another people's work), your score will be zero.
- 5.2** The application must fulfil the following criteria in addition to your design:
- It must be tested to work on emulator or device
  - Graphics used must be original
  - All errors must be resolved

## 6. Project Assessment Plan

This project is worth **40%** of the Cross Platform Mobile Application Development module. The delivery and assessment plan are as follow:

**Week 11** – Submission of Project Proposal Writeup (**3%**)

**Week 16** – Milestone check – with program structure, and code outline defined.

**Week 18** – Demonstration and Submission of completed project (**37%**) – include code walk through\*

**\* a. You must complete the assessment by week 18. Failure to do so will result in a penalty of 5 marks per day.**

**b. Code walkthrough looks at liberal use of comments to enhance readability of codes, elegance of algorithms & 'cleanliness' of codes. 'Cleanliness' of code refers to removal of variables, functions that are not utilised among others.**

**c. Refer to Annex – Project Rubric**

Besides meeting requirement of the project, students will be rewarded for

- a. Pleasant User Experience (e.g., no one-way navigation, not seemingly 'hang' situation, pleasant colour scheme and choice of text style)
- b. Innovative User Interface (e.g., intuitive and informative interface)
- c. Idiot Proof Design (e.g., optimised layout of UI components)

## 7. Project Proposal (3%)

You must submit a project proposal to describe your project. The writeup should include the following sections:

|                                                                                                                                                                                                                                                                                                                                                         |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <p><b>Name:</b></p>                                                                                                                                                                                                                                                                                                                                     | <p><b>Group:</b></p> |
| <p><b>A. Project Title</b></p> <p>Please give a meaningful name for your application.</p>                                                                                                                                                                                                                                                               |                      |
| <p><b>B. Background</b></p> <p>Why do you want to build this mobile application?</p>                                                                                                                                                                                                                                                                    |                      |
| <p><b>C. Objectives</b></p> <p>What do you want to achieve with this mobile application?</p>                                                                                                                                                                                                                                                            |                      |
| <p><b>D. Key Functional Features</b></p> <p>Write down clearly what are the things that user can perform with your proposed application (<b><u>minimum 4 functional features</u></b>). You may attach use case &amp; UI design diagrams.</p> <p>For clarity and fairness, <u>user login, registration and logout are considered as one feature.</u></p> |                      |
| <p><b>E. Data Source &amp; Data Storage</b></p> <p>State the source of data whether it's hardcoded or live JSON dataset.<br/>State the data storage method such as Firestore, or any DBMS.</p>                                                                                                                                                          |                      |
| <p><b>F. State Management</b></p> <p>State the State Management approach, e.g., Provider or GetX.</p>                                                                                                                                                                                                                                                   |                      |
| <p><b>G. References</b></p> <p>If you have done any research to support your proposal, feel free to put the links in this section.</p>                                                                                                                                                                                                                  |                      |

Please send your proposal (save as: '**L#\_CPMAD\_Proposal\_YourName.docx**') via **email** (**Ang\_Beng\_Hoe@nyp.edu.sg**) or **WhatsApp by Week-11** for approval. Do not proceed with the project implementation until your proposal is approved. However, you may do the initial project setup while waiting for the approval.

## 8. Project Submission

The mode of submission for project codes is via GitHub repository only.

- You must create a GitHub private repository for code versioning. You can read up on [Version Control Best Practices](#) for more information on version controlling.
- You can also link Visual Studio Code with GitHub so that you can just commit and update codes using VS Code. Find out more by watching [this video](#) or following [this link](#).
- Please add your lecturer as a collaborator using his/her email.

*\* You must complete the **submission by week 18 - Friday**. Failure to do so will result in a **penalty of 5 marks per day**.*